5.15 Construction of a Rectangle

Let us now construct a rectangle with the given measurements

(i) length and breadth

(ii) a side and a diagonal

5.15.1 Construction of a rectangle when its length and breadth are given

Example 5.38

Construct a rectangle BEAN with BE = 5 cm and BN = 3 cm. Also find its area.

Solution:

Given: BE = 5 cm and BN = 3 cm

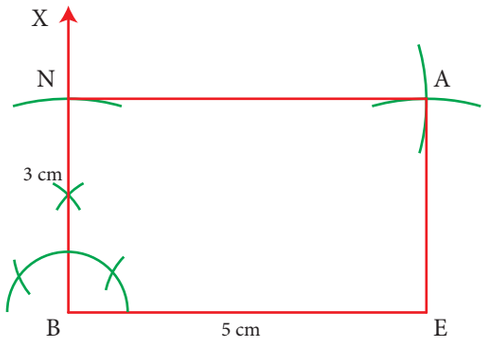
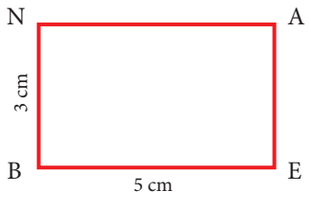


Fig. 5.57



Rough diagram

Steps:

1. Draw a line segment BE = 5 cm.
2. At B, construct $BX \perp BE$.
3. With B as centre, draw an arc of radius 3 cm and let it cut BX at N.
4. With E and N as centres, draw arcs of radii 3 cm and 5 cm respectively and let them cut at A.
5. Join EA and NA.
6. BEAN is the required rectangle.

Calculation of Area:

Area of rectangle BEAN $= l \times b \, sq.units$

$= 5 \times 3 = 15 \, sq.cm$

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